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Specification for
**Graphical symbols for
general engineering**

Part 1. Piping systems and plant

Spécification des symboles graphiques pour la construction mécanique générale
Partie 1. Canalisations et équipements

Spezifikation für Graphische Symbole für allgemeinen Maschinenbau
Teil 1. Rohrsysteme und Anlagen

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Foreword

For many years BSI has published separate specifications for graphical symbols used on engineering diagrams relating to individual technical disciplines; this has inevitably led to a few inconsistencies between standards. In this revision of BS 1553 : Part 1 : 1949, which has been prepared under the authority of the Mechanical and the Chemical Standards Committees, an attempt has been made to eliminate such differences and produce a standard which specifies symbols which will be common to heating and ventilating installations and to process plants of all types. To accomplish this

aim BS 1553 : Part 4 : 1956 'Graphical symbols for heating and ventilating installations' and BS 974 : 1953 'Symbols for use on flow diagrams of chemical and petroleum plant', have also been revised and incorporated into this standard. BS 1553 : Part 4 and BS 974 have accordingly been withdrawn.

It is intended that the symbols specified in this standard should be applied in accordance with the practice recommended in BS 5070.

British Standard Specification for

Graphical symbols for general engineering

Part 1. Piping systems and plant

1. Scope

This Part of BS 1553 specifies graphical symbols for use in the creation of flow and piping diagrams for process plant and heating and ventilating installations. It is intended that diagrams employing these symbols should be drawn in accordance with the practice recommended in BS 5070.

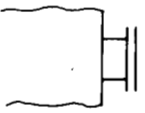
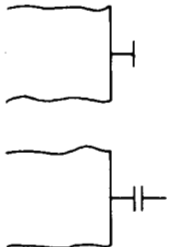
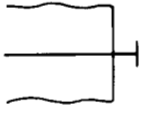
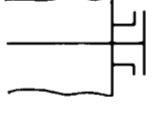
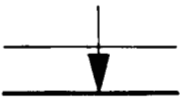
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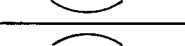
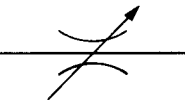
The titles of the standards publications referred to in this standard are listed on the inside back cover.

3. Symbols (or elements of symbols) used in conjunction with other symbols

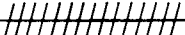
3.1 General

No.	Description	Symbol
3.1.1	Mechanical linkage	
3.1.2	Weight device	
3.1.3	Electrical device	
3.1.4	Vibratory or loading device (any type)	
3.1.5	Spray device	
3.1.6	Rotary movement	
3.1.7	Stirring device	
3.1.8	Fan	

No.	Description	Symbol
3.1.9	Access point	
3.1.10	Equipment branch: general symbol NOTE. The upper representation does not necessarily imply a flange, merely the termination point. Where a breakable connection is required the branch/pipe would be as shown in the lower symbol.	
3.1.11	Equipment penetration (fixed)	
3.1.12	Equipment penetration (removable)	
3.1.13	Boundary line	
3.1.14	Point of change (See 4.1.12 for specific examples)	
3.1.15	Discharge to atmosphere	

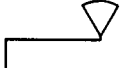
No.	Description	Symbol
3.1.16	Intake from atmosphere	
3.1.17	Trap: release or retention	
3.1.18	Flow restrictor (fixed)	
3.1.19	Flow restrictor (variable)	

3.2 Pipelines

3.2.1	Pipeline (major flow line)	
3.2.2	Pipeline (minor flow line)	
3.2.3	Pipeline (existing)	
3.2.4	Pipeline (undground, or concealed)	
3.2.5	Existing pipeline (to be removed)	
3.2.6	Pipeline (future)	

3.3 Valves: actuating methods

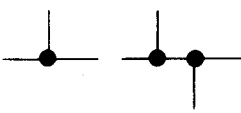
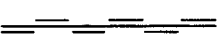
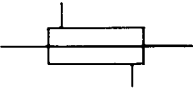
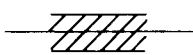
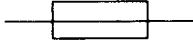
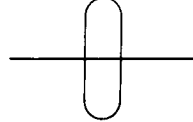
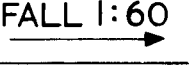
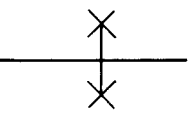
3.3.1	Manual isolation	
3.3.2	Power signal	

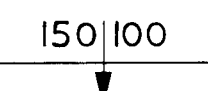
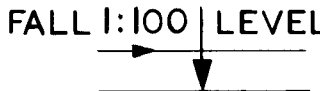
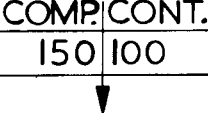
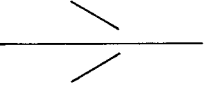
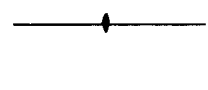
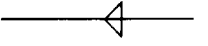
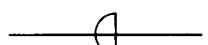
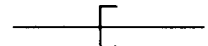
3.3.3	Statically loaded	 i.e. symbol 3.1.4
3.3.4	Spring loaded	
3.3.5	Weight loaded	
3.3.6	Quick opening	
3.3.7	Quick closing	
3.3.8	Float control	
3.3.9	Dash pot checked	
3.3.10	Piston	
3.3.11	Diaphragm	
3.3.12	Solenoid	
3.3.13	Automatic actuating element	
3.3.14	Automatic actuating element, with integral manual actuating element	
3.3.15	Manual actuating element	

No.	Description	Symbol
3.3.16	Motor	

4. Basic functional symbols

4.1 Pipelines

4.1.1	Crossing (unconnected) (See appendix A)	
4.1.2	Junction (connected) (see appendix A)	
4.1.3	Heated or cooled NOTE. Heating or cooling medium to be annotated adjacent to symbol	
4.1.4	Jacketed NOTE. Heating or cooling medium to be annotated adjacent to symbol	
4.1.5	Lagged	
4.1.6	Sleeved NOTE. Fluid or fill of the annulus to be annotated adjacent to symbol	
4.1.7	Provision for flexibility (See 4.7 for specific methods)	
4.1.8	Indication of flow direction NOTE. Included angle of arrow = 30°	
4.1.9	Indication of pipe size ø150 mm example	
4.1.10	Indication of fall (or rise)	
4.1.11	Indication of support points (See 4.4 for specific types of supports and hangers)	

4.1.12	Indication of point of change (i.e. size, specification, level, responsibility, etc. added to symbol 3.1.14) NOTE. Changes in conditions should embrace the arrow Examples: Responsibility change Pipe bore change Change in fall	  
4.1.13	Coincident point of change	
4.1.14	Pipe bore change (unspecified)	
4.2 Joints		
4.2.1	Butt welded NOTE. The proportions of the weld should be thus: 4 x line thickness	
4.2.2	Soldered or solvent welded	
4.2.3	Screwed	
4.2.4	Compression	
4.2.5	Sleeve: BS 534	

No.	Description	Symbol
4.2.6	Socket welded: BS 3799	
4.2.7	Socket and spigot	
4.2.8	Swivel	
4.2.9	Flanged and bolted	
4.2.10	Electrically bonded	
4.2.11	Electrically insulated	

4.3 End closures

4.3.1	End cap: general symbol (method unspecified) (Annotate as 'detachable', when appropriate)	
4.3.2	End cap: butt welded	
4.3.3	End cap: screwed	
4.3.4	End cap: fillet welded	
4.3.5	End screwed and plugged	
4.3.6	End flanged and bolted	

No.	Description	Symbol
4.3.7	End closure: quick release	
4.3.8	End socket and spigot	

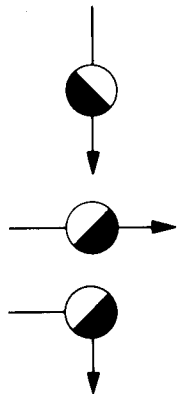
4.4 Supports and hangers

4.4.1	Support: basic symbol (Use symbol 4.1.11 when a detailed support or hanger is not required)	See 4.1.11
4.4.2	Simple support	
4.4.3	Resilient support	
4.4.4	Adjustable support	
4.4.5	Roller support	
4.4.6	Simple hanger	
4.4.7	Resilient hanger	
4.4.8	Adjustable hanger	
4.4.9	Constant load hanger	
4.4.10	Guide	

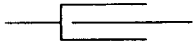
No.	Description	Symbol
4.4.11	Anchor	
4.5 Valves		
4.5.1	In-line (any type or pattern)	
4.5.2	In-line (simple screw-down)	
4.5.3	Angle (simple screw-down)	
4.5.4	Relief (in-line)	
4.5.5	Relief (angle, pressure)	
4.5.6	Relief (angle, vacuum)	
4.5.7	Check (non-return or reflux)	
4.5.8	Check (non-return or reflux) screw-down	
4.5.9	3-way	
4.5.10	4-way	

No.	Description	Symbol
4.5.11	In-line (float operated)	
4.5.12	Flanged or alternatively	
4.5.13	Butterfly	
4.5.14	Globe	
4.5.15	Ball	
4.5.16	Wedge gate	
4.5.17	Parallel slide	
4.5.18	Needle	
4.5.19	Diaphragm	
4.5.20	Plug Annotate type of port adjacent to symbol i.e. T (Tee port) L (L port)	
4.5.21	Plug 3-way Annotate type of port adjacent to symbol i.e. T (Tee port) L (L port)	

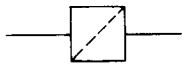
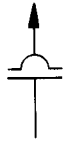
4.6 Trap functions

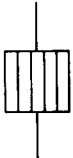
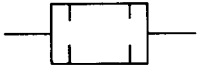
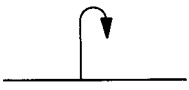
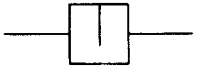
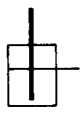
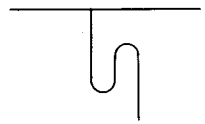
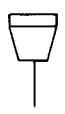
No.	Description	Symbol
4.6.1	Trap drain, e.g. condensate release	
4.6.2	Trapped vent, e.g. automatic air valve	

4.7 Pipeline flexibility

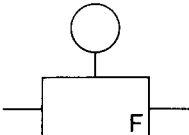
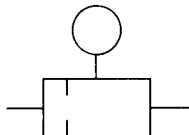
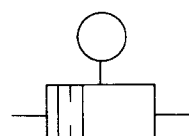
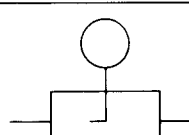
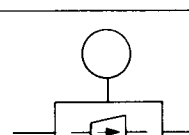
4.7.1	Bellows	
4.7.2	Sleeve extension	
4.7.3	Expansion loop	
4.7.4	Flexible hose, flanged example	

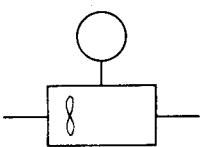
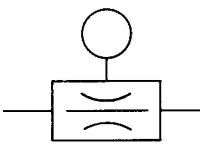
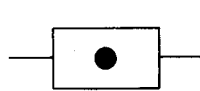
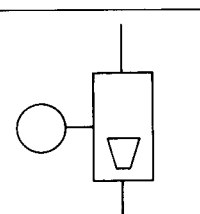
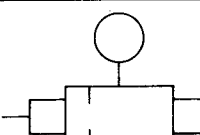
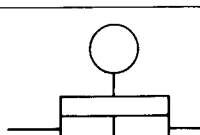
4.8 Pipeline features and general equipment

4.8.1	Strainer or filter	
4.8.2	Bursting disc	

No.	Description	Symbol
4.8.3	Flame arrester	
4.8.4	Silencer	
4.8.5	Tundish NOTE. Arrow to be added when discharge is to atmosphere	
4.8.6	Open vent	
4.8.7	Separator	
4.8.8	Bell mouth	
4.8.9	Liquid seal, open	
4.8.10	Liquid seal, closed	
4.8.11	Syphon drain	
4.8.12	Hydrant	
4.8.13	Exhaust head	

4.9 Sensing elements for measurement and control

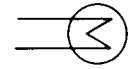
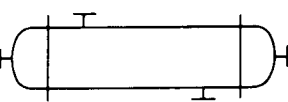
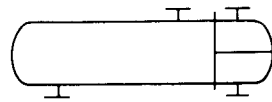
No.	Description	Symbol
4.9.1	Point of measurement	
4.9.2	Meter (any type) NOTE. Letter F indicates flow measurement	
4.9.3	Orifice plate	
4.9.4	Orifice carrier	
4.9.5	Pitot tube	
4.9.6	Turbine or propellor	

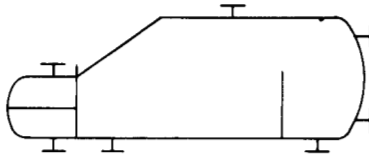
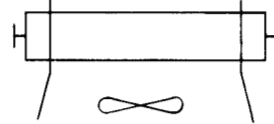
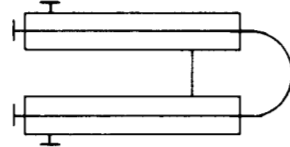
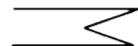
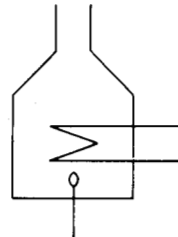
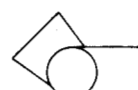
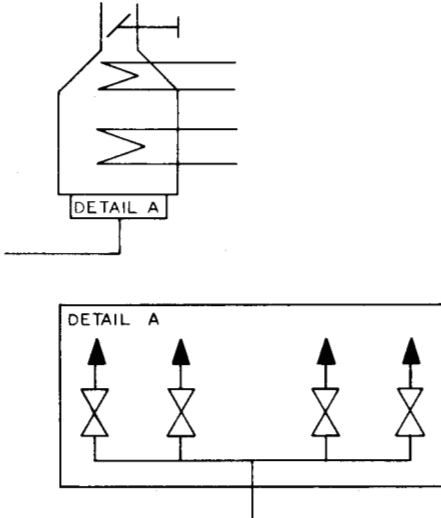
No.	Description	Symbol
4.9.7	Rotary displacement	
4.9.8	Venturi	
4.9.9	Sight flow indicator	
4.9.10	Variable area meter	
4.9.11	Calibrated pipe run	
4.9.12	Diaphragm meter	

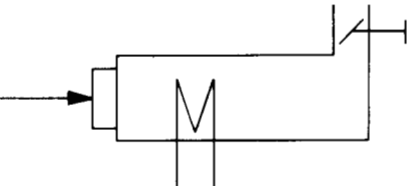
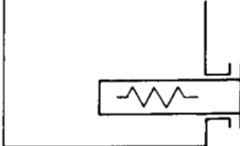
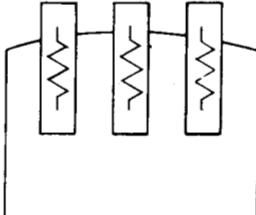
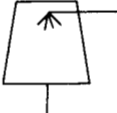
NOTE. For letter codes within the instrument circle and for detailed measurement and control symbols see BS 1646.

No.	Description	Symbol
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5. Basic and developed symbols for plant and equipment**5.1 Heat transfer equipment**

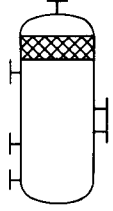
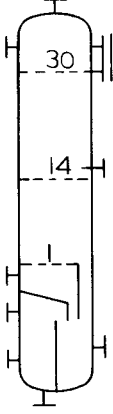
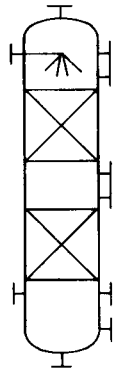
5.1.1	Heat exchanger (basic symbols) Alternative:	 
5.1.2	Shell and tube: fixed tube sheet	
5.1.3	Shell and tube: U tube or floating head	

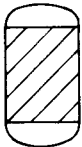
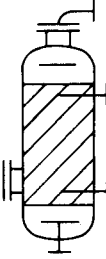
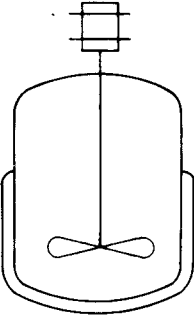
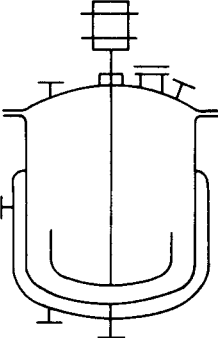
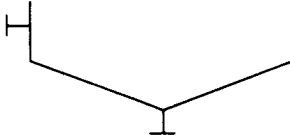
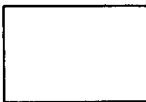
No.	Description	Symbol
5.1.4	Shell and tube: kettle reboiler	
5.1.5	Air-blown cooler	
5.1.6	Plate type	
5.1.7	Double pipe type	
5.1.8	Heating/cooling coil (basic symbol)	
5.1.9	Fired heater/boiler (basic symbol)	
5.1.10	Automatic stoker	
5.1.11	Upshot heater Where complex burners are employed the 'burner block' may be detailed elsewhere on the drawing, thus:	

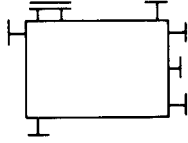
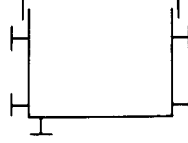
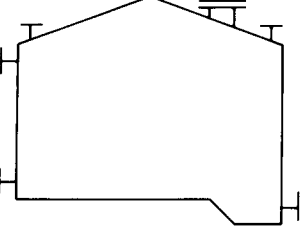
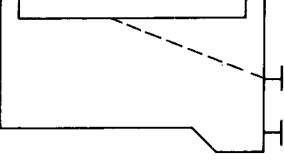
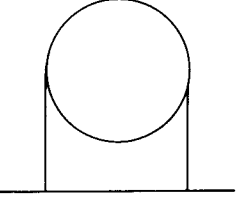
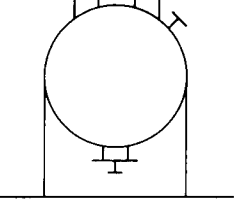
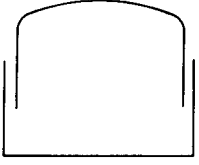
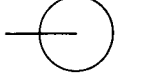
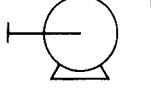
No.	Description	Symbol
5.1.12	Cabin heater	
5.1.13	Fire tube boiler	
5.1.14	Electrical heater (basic symbol)	
5.1.15	Oil burner	
5.1.16	Gas burner	
5.1.17	Tank heater	
5.1.18	Electric furnace	
5.1.19	Cooling tower (basic symbol)	
5.1.20	Cooling tower: natural draft	

No.	Description	Symbol
5.1.21	Cooling tower: forced/induced draft (Fans to be included as appropriate)	
5.1.22	Water-cooled condenser	
5.1.23	Air-cooled condenser, natural/forced draft Add fan 3.1.8 if required	
5.1.24	Evaporative condenser	
5.1.25	Unit humidifier	
5.1.26	Water chilling evaporator	
5.1.27	Air cooling evaporator	
5.1.28	Oil separator	
5.1.29	Extractor hood, open type	
5.1.30	Extractor hood, slot type	

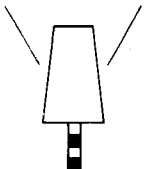
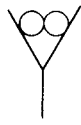
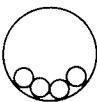
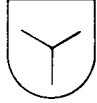
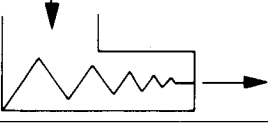
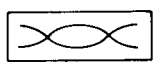
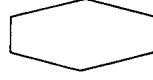
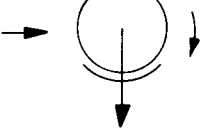
5.2 Vessels and tanks

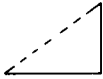
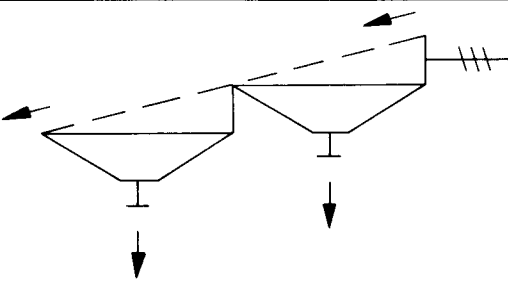
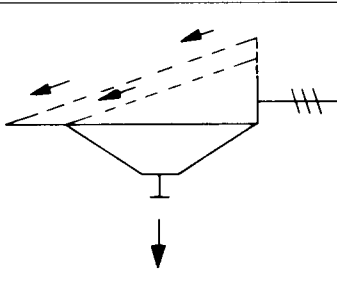
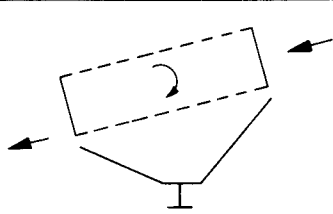
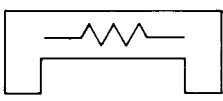
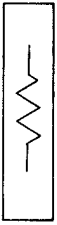
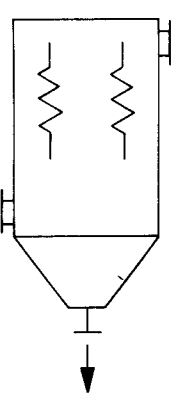
No.	Description	Symbol
5.2.1	Drum or simple pressure vessel (basic symbol)	
5.2.2	Knock-out drum (with demister pad)	
5.2.3	Tray column (basic symbol)	
5.2.4	Tray column Trays should be numbered from the bottom; at least the first and the last should be shown. Intermediate trays should be included and numbered where they are significant.	
5.2.5	Fluid contacting vessel (basic symbol)	
5.2.6	Fluid contacting vessel Support grids and distribution details may be shown	

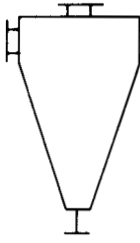
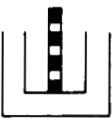
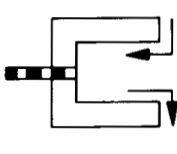
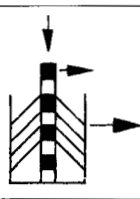
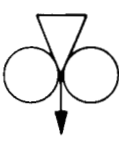
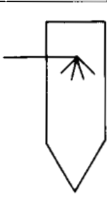
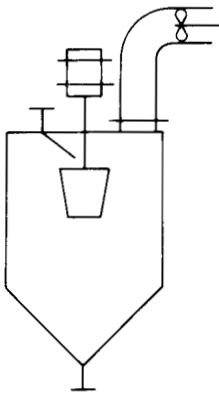
No.	Description	Symbol
5.2.7	Reaction or absorption vessel (basic symbol)	
5.2.8	Reaction or absorption vessel Where it is necessary to show more than one layer of material alternate hatching should be used	
5.2.9	Autoclave (basic symbol)	
5.2.10	Autoclave	
5.2.11	Open tank (basic symbol)	
5.2.12	Open tank	
5.2.13	Clarifier or setting tank	
5.2.14	Closed tank (basic symbol)	

No.	Description	Symbol
5.2.15	Sealed tank	
5.2.16	Covered tank	
5.2.17	Tank with fixed roof (with draw-off sump)	
5.2.18	Tank with floating roof (with roof drain)	
5.2.19	Storage sphere (basic symbol)	
5.2.20	Storage sphere	
5.2.21	Gas holder (basic symbol for all types)	
5.3 Pumps and compressors		
5.3.1	Rotary pump, fan or simple compressor (basic symbol)	
5.3.2	Centrifugal pump or centrifugal fan	

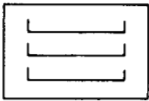
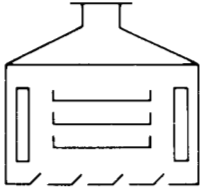
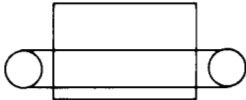
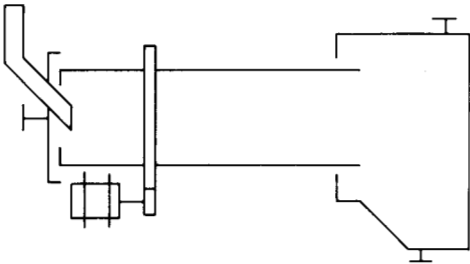
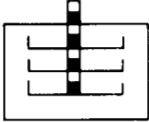
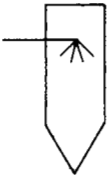
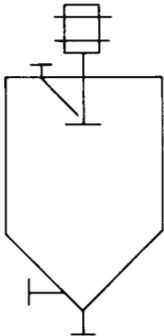
No.	Description	Symbol
5.3.3	Centrifugal pump (submerged suction)	
5.3.4	Positive displacement rotary pump or rotary compressor	
5.3.5	Positive displacement pump (reciprocating)	
5.3.6	Axial flow fan	
5.3.7	Compressor: centrifugal/axial flow (basic symbol)	
5.3.8	Compressor: centrifugal/axial flow	
5.3.9	Compressor: reciprocating (basic symbol)	
5.3.10	Two-stage, electric motor driven reciprocating compressor	
5.3.11	Ejector/injector (basic symbol)	
5.4 Solids handling		
5.4.1	Size reduction (basic symbol)	

No.	Description	Symbol
5.4.2	Breaker: gyratory	
5.4.3	Roll crusher	
5.4.4	Hammer crusher	
5.4.5	Pulverizer: ball mill	
5.4.6	Mixing (basic symbol)	
5.4.7	Kneader	
5.4.8	Ribbon blender	
5.4.9	Double cone blender	
5.4.10	Filter (basic symbol, simple batch)	
5.4.11	Filter press (basic symbol)	
5.4.12	Rotary filter (basic symbol)	
5.4.13	Rotary filter, film drier or flaker	

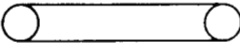
No.	Description	Symbol
5.4.14	Screen (basic symbol)	
5.4.15	Vibrated single-deck screen	
5.4.16	Vibrated multi-deck screen	
5.4.17	Rotary screen	
5.4.18	Electro-magnet (basic symbol)	
5.4.19	Electrostatic precipitator (basic symbol)	
5.4.20	Electrostatic precipitator	

No.	Description	Symbol
5.4.21	Cyclone and hydroclone (basic symbol)	
5.4.22	Cyclone and hydroclone	
5.4.23	Centrifuge (basic symbol)	
5.4.24	Centrifuge: horizontal peeler type	
5.4.25	Centrifuge: disc bowl type	
5.5 Shaping and forming		
5.5.1	Briquetting machine (basic symbol)	
5.5.2	Prill tower (basic symbol)	
5.5.3	Prill tower	

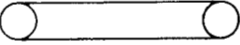
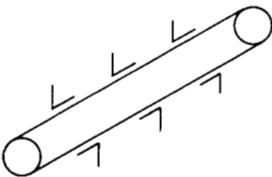
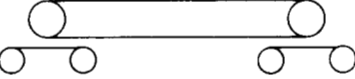
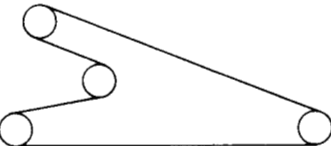
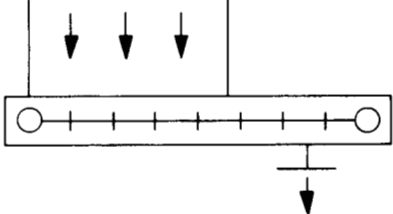
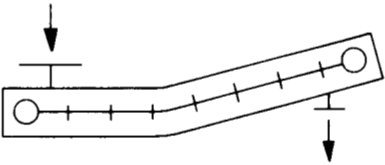
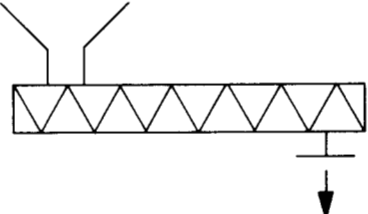
5.6 Drying

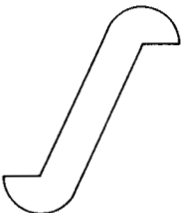
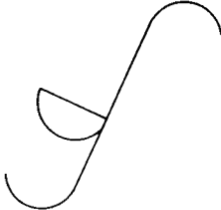
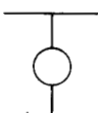
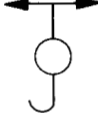
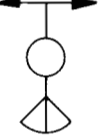
No.	Description	Symbol
5.6.1	Batch tray dryer (basic symbol)	
5.6.2	Drying oven	
5.6.3	Belt drier (basic symbol)	
5.6.4	Rotary drier (basic symbol)	
5.6.5	Rotary kiln	
5.6.6	Hot gas drier with trays (basic symbol)	
5.6.7	Spray drier (basic symbol)	
5.6.8	Spray drier	

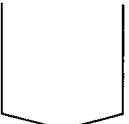
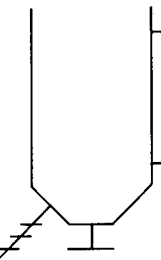
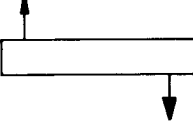
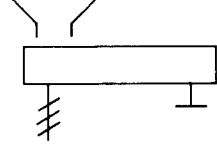
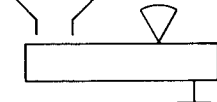
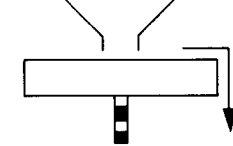
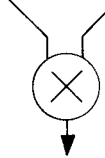
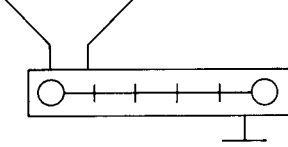
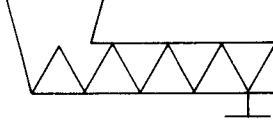
5.7 Materials handling

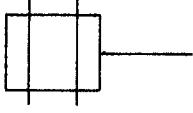
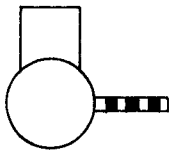
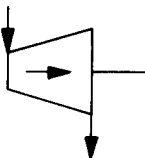
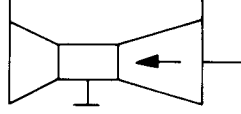
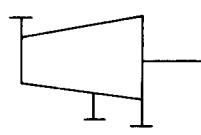
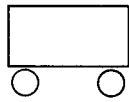
No.	Description	Symbol
5.7.1	Conveyor (basic symbol)	

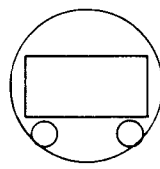
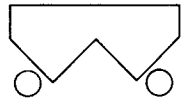
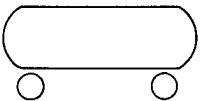
NOTE. It is thought that for the elementary flow diagram, the basic symbol would be the only symbol necessary to illustrate a conveyor, mainly because the specification of the equipment would rarely be known at that stage. It is therefore recommended that wherever possible, the basic symbol should be used for a conveyor of any type. However, if it is felt that some elaboration must be given, the following symbols are shown to demonstrate the development of the basic symbol.

5.7.2	Belt conveyor	
5.7.3	Inclined conveyor (with chevrons)	
5.7.4	Boom loader	
5.7.5	Shuttle conveyor	
5.7.6	Tripper conveyor	
5.7.7	Scraper conveyor (en masse)	
5.7.8	Inclined scraper conveyor	
5.7.9	Screw conveyor	

No.	Description	Symbol
5.7.10	Elevator (basic symbol)	
5.7.11	Skip hoist (basic symbol)	
5.7.12	Overhead conveyor (basic symbol)	
5.7.13	Overhead conveyor with hooks	
5.7.14	Overhead conveyor or ropeway with buckets or carriers	
5.7.15	Hoist (basic symbol)	
5.7.16	Fixed hoist with hook	
5.7.17	Travelling hoist with hook	
5.7.18	Travelling hoist with grab	
5.7.19	Weighing machine (Refer to symbol 3.1.2)	
5.7.20	Weighbridge (Refer to symbol 3.1.2)	

No.	Description	Symbol
5.7.21	Hopper bunker or silo (basic symbol)	
5.7.22	Bunker with vibrator	
5.7.23	Feeder (basic symbol)	
5.7.24	Vibrator feeder	
5.7.25	Weigh feeder	
5.7.26	Rotary table feeder	
5.7.27	Rotary valve feeder	
5.7.28	Scraper feeder	
5.7.29	Screw feeder	

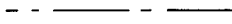
No.	Description	Symbol
5.8 Prime movers		
5.8.1	Electric motor (basic symbol)	
5.8.2	Reciprocating engine (basic symbol)	
5.8.3	Turbine (basic symbol)	
5.8.4	Gas turbine engine	
5.8.5	Steam turbine with intermediate pass-out	
5.9 Trucking		
5.9.1	Wagon, tub or bogey	

No.	Description	Symbol
5.9.2	Wagon, rotary tipper	
5.9.3	Wagon, bottom hopper	
5.9.4	Road or rail tanker	

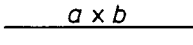
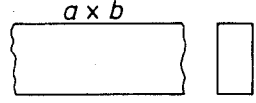
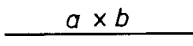
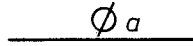
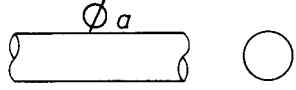
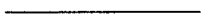
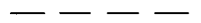
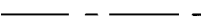
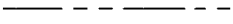
6. Heating and ventilating installations

NOTE. Refer to appendices B and C for examples of the use of symbols for such installations.

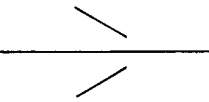
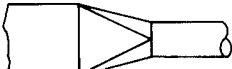
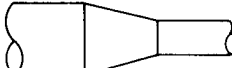
6.1 Pipelines

6.1.1	Pipe in front of (or above) section	
6.1.2	Pipe in roof or above ceiling (a specific case for plans only)	
6.1.3	View of pipe perpendicular to the plane of the drawing	

6.2 Ducts (trunks)

No.	Description	Symbol	
		Single line convention	Double line convention
6.2.1	Duct, rectangular NOTE. The size of the rectangular duct should be indicated by its dimensions $a \times b$ where a is the dimension seen in the view drawn		
6.2.2	Duct, rectangular-oval NOTE. The size of the rectangular oval duct should be indicated by its dimensions $a \times b$, where a is the dimension in the view drawn.		
6.2.3	Duct, circular NOTE. The diameter of the circular duct should be indicated by ϕa		
6.2.4	Indication of direction of flow, inlet or supply.	 i.e. as symbol 4.1.8	
6.2.5	Indication of direction of flow, outlet or exhaust.		
6.2.6	Duct at low level or visible at section	 i.e. as symbol 3.2.1	
6.2.7	Duct concealed at plan or section	 i.e. as symbol 3.2.4	
6.2.8	Duct at high level on plan, or in front of (or above) section	 i.e. as symbol 6.1.1	
6.2.9	Duct in roof or above ceiling (a specific case for plans only)	 i.e. as symbol 6.1.2	

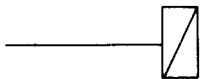
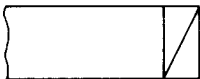
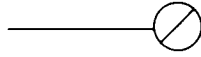
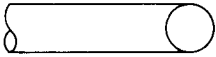
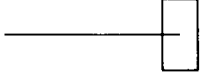
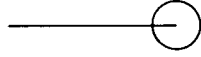
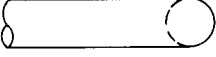
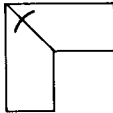
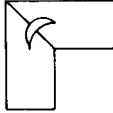
6.3 Changes of section and/or size of duct (trunk)

No.	Description	Symbol	
		Single line convention	Double line convention
6.3.1	Rectangular to rectangular		 (See note)
6.3.2	Rectangular to circular		 (See note)
6.3.3	Circular to circular		 (See note)

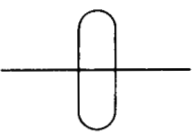
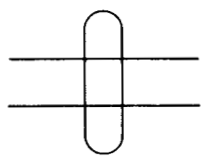
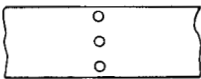
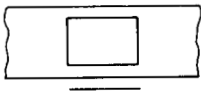
i.e. as symbol 4.1.14

NOTE. Applicable dimensions should be annotated.

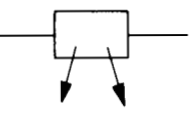
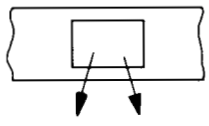
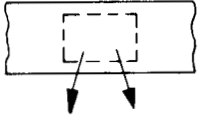
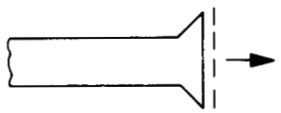
6.4 Duct (trunk) bends: representation and design

No.	Description	Symbol	
		Single line representation	Double line representation
6.4.1	Rectangular duct bend (visible section)		
6.4.2	Circular duct bend visible (visible section)		
6.4.3	Rectangular duct bend (hidden section)		
6.4.4	Circular duct bend (hidden section)		
6.4.5	Bend with internal splitters or vanes	not applicable	Number splitters to be annotated on drawing 
6.4.6	Mitre bend with internal vanes		
6.4.7	Mitre bend with internal aerofoil vanes		

6.5 Duct (trunk) fittings

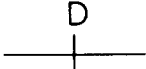
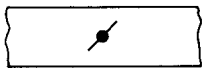
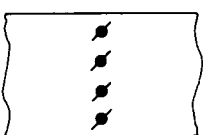
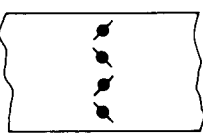
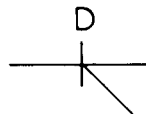
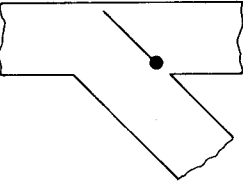
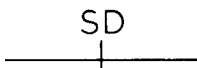
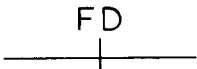
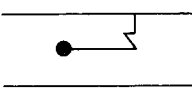
No.	Description	Symbol	
		Single line representation	Double line representation
6.5.1	Flexible duct	 i.e. as symbol 4.7.4	
6.5.2	Flexible connection	 i.e. as symbol 4.1.7	
6.5.3	Test holes	Test holes 	
6.5.4	Access door or cover	 AD	

6.6 Grilles and diffusers

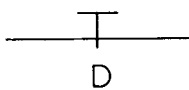
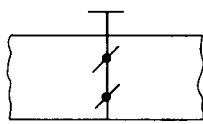
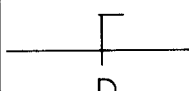
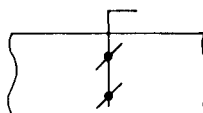
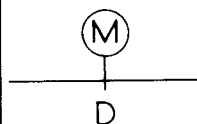
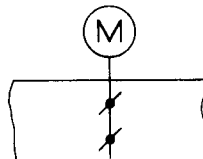
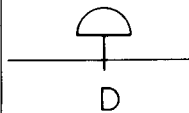
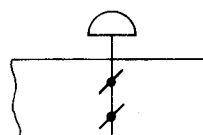
6.6.1	Grille, diffuser: facing (or near side)		
6.6.2	Grille, diffuser: remote (or far side)		
6.6.3	Grille, diffuser: edge view		
6.6.4	Grille, diffuser: on end of duct		

NOTE. The arrows in these symbols denote supply from the duct. The direction and configuration of the arrows define the function of the equipment. Arrows 6.2.4 and 6.2.5 should be used, as appropriate.

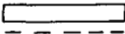
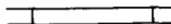
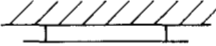
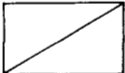
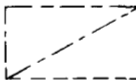
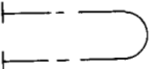
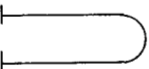
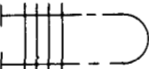
6.7 Dampers

No.	Description	Symbol	
		Single line convention	Double line convention
6.7.1	Single leaf damper		
6.7.2	Multileaf, parallel blade damper		
6.7.3	Multileaf, opposed blade damper		
6.7.4	Deflecting damper		
6.7.5	Sliding damper		
6.7.6	Fire damper		Annotate direction of flow 

6.8 Damper actuation

6.8.1	Manual Reflecting symbol 3.3.1		
6.8.2	Power: unspecified Reflecting symbol 3.3.2		
6.8.3	Power: motor Reflecting symbol 3.3.16		
6.8.4	Power: pneumatic (diaphragm) Reflecting symbol 3.3.11		

6.9 Heat exchange equipment

No.	Description	Symbol
6.9.1	Radiator	
6.9.2	Convactor, natural	
6.9.3	Convactor, fan	
6.9.4	Radiant panel, industrial, single (plan view)	
6.9.5	Radiant panel, industrial, double (plan view)	
6.9.6	Radiant panel, wall mounted (plan view)	
6.9.7	Radiant panel, ceiling mounted (face view)	
6.9.8	Radiant panel, embedded in floor or wall (face view)	
6.9.9	Radiant panel, embedded in ceiling or wall, behind section (face view)	
6.9.10	Pipe coils, plain: in front of, or above, section	
6.9.11	Pipe coils, plain: visible at section	
6.9.12	Pipe coils, finned: in front of, or above, section	

No.	Description	Symbol
6.9.13	Pipe coils, finned: visible at section	
6.9.14	Towel rail	
6.9.15	Radiant strip: in front of, or above, section	
6.9.16	Heater/cooler unit, horizontal type Example: heater/cooler unit	
6.9.17	Heater/cooler unit, downward type Example: heater unit	
6.9.18	Heater/cooler unit, floor mounted-type	

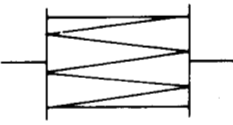
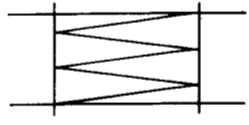
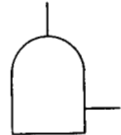
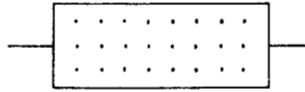
NOTE. Annotate to indicate function, as follows.

Heater: H or +

Cooler: C or -

6.10 Air conditioning and ventilation equipment

No.	Description	Symbol	
		Single line convention	Double line convention
6.10.1	Propeller fan	 i.e. as symbol 3.1.8	
6.10.2	Centrifugal fan	 i.e. as symbol 5.3.1	
6.10.3	Moisture eliminator		
6.10.4	Attenuator (silencer)	 i.e. as symbol 4.8.4	

No.	Description	Symbol	
		Single line convention	Double line convention
6.10.5	Air filter		
6.10.6	Refrigerant receiver		Not applicable
6.10.7	Drier		Not applicable

Appendix A

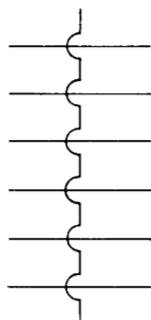
Representation of flow lines crossing (connected and unconnected)

There are several variants in the symbolic representation of flow lines (pipelines) crossing, connected and unconnected, as depicted on engineering diagrams and engineering drawings. In order to appreciate the reasoning behind some of the alternatives it is necessary first to state the differences between the requirements of an engineering diagram and the requirements of an engineering drawing.

A.1 Diagram. An engineering diagram is usually prepared in single-plane form for the purpose of showing in the simplest possible manner the principal inter-relationships of elements in a system and how the system operates. It may show in full the functioning of a circuit, a process sequence or an installation, symbolically depicting all the essential units, parts and connections to show the layout in the clearest possible manner but without regard for the physical disposition of such features. Apart from giving such information as pipe size, linear dimensions are rarely used on engineering diagrams because they are not normally the proper vehicle for conveying information on precise physical or geographical relationships.

On an engineering diagram, flow lines crossing but unconnected are represented thus,  while flow lines crossing and connected are represented by  or . Flow lines connected but without crossing are represented thus, .

In past years alternative methods employing loops (a) and broken lines (b) have been used but on complex diagrams involving closely spaced, parallel flow lines these methods are not practicable



(a) deprecated



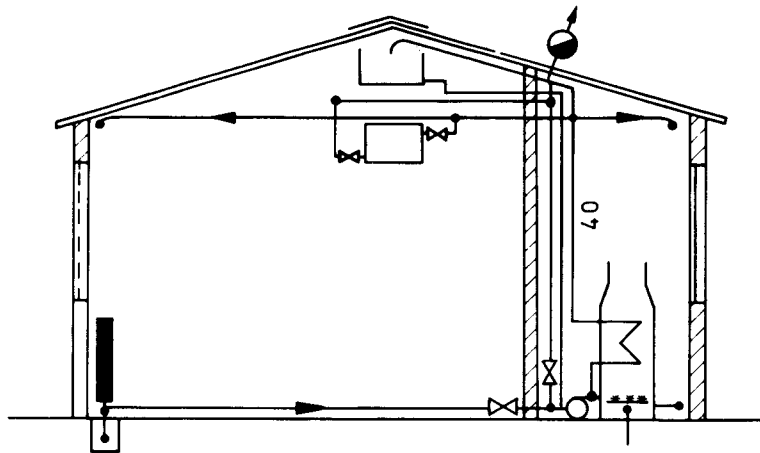
(b) deprecated

A.2 Drawing. An engineering drawing is used to show how a part is to be made, or how parts are assembled and installed relative to other equipment. The physical relationships of components and equipments and their disposition are important and therefore precise dimensions and locations are fully stated.

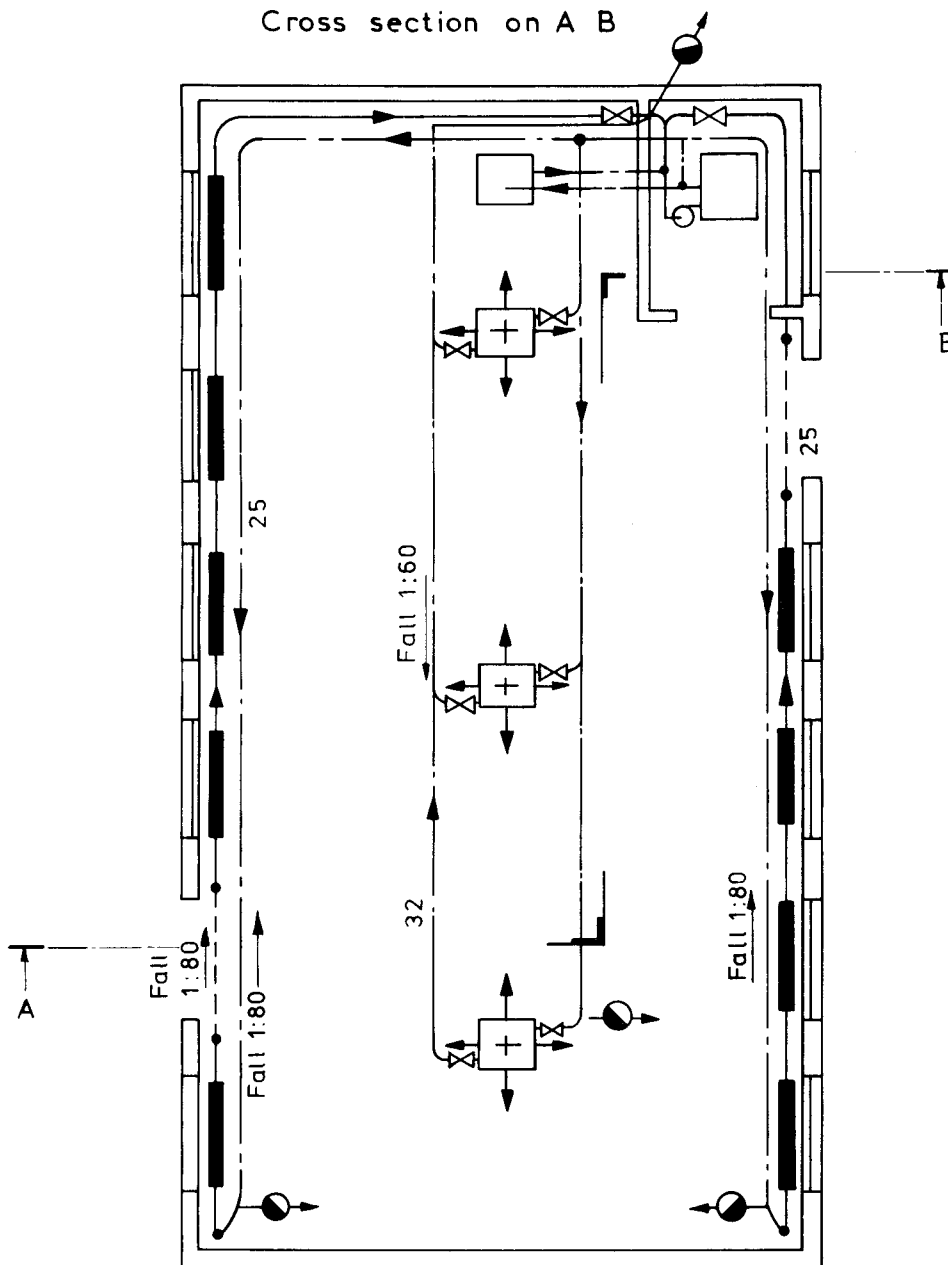
The installation of a heating, ventilating or refrigeration plant within a building or other structure, e.g. a ship, will be shown on a drawing rather than on a diagram. Contrary to the single-plane presentation of a diagram, a drawing is of multi-plane form and pipelines may be indicated normal to the plane of the drawing itself. In an engineering drawing a dot on a line thus,  would indicate that a pipeline rises or falls vertically from the plane of the drawing and is connected to another pipe running horizontally. Hence, it is necessary always to realize the distinct purpose of the engineering drawing and the engineering diagram and, by so doing, to avoid confusion between methods of representation which at first sight are similar but which, in their separate contexts, have very different interpretations. Because this is an accepted drawing discipline, misunderstandings are in reality very rare and their possibility can be further reduced by the use of additional elevations on the drawing.

Appendix B

Example of the use of symbols on a drawing of a low temperature hot water heating installation



Cross section on A B

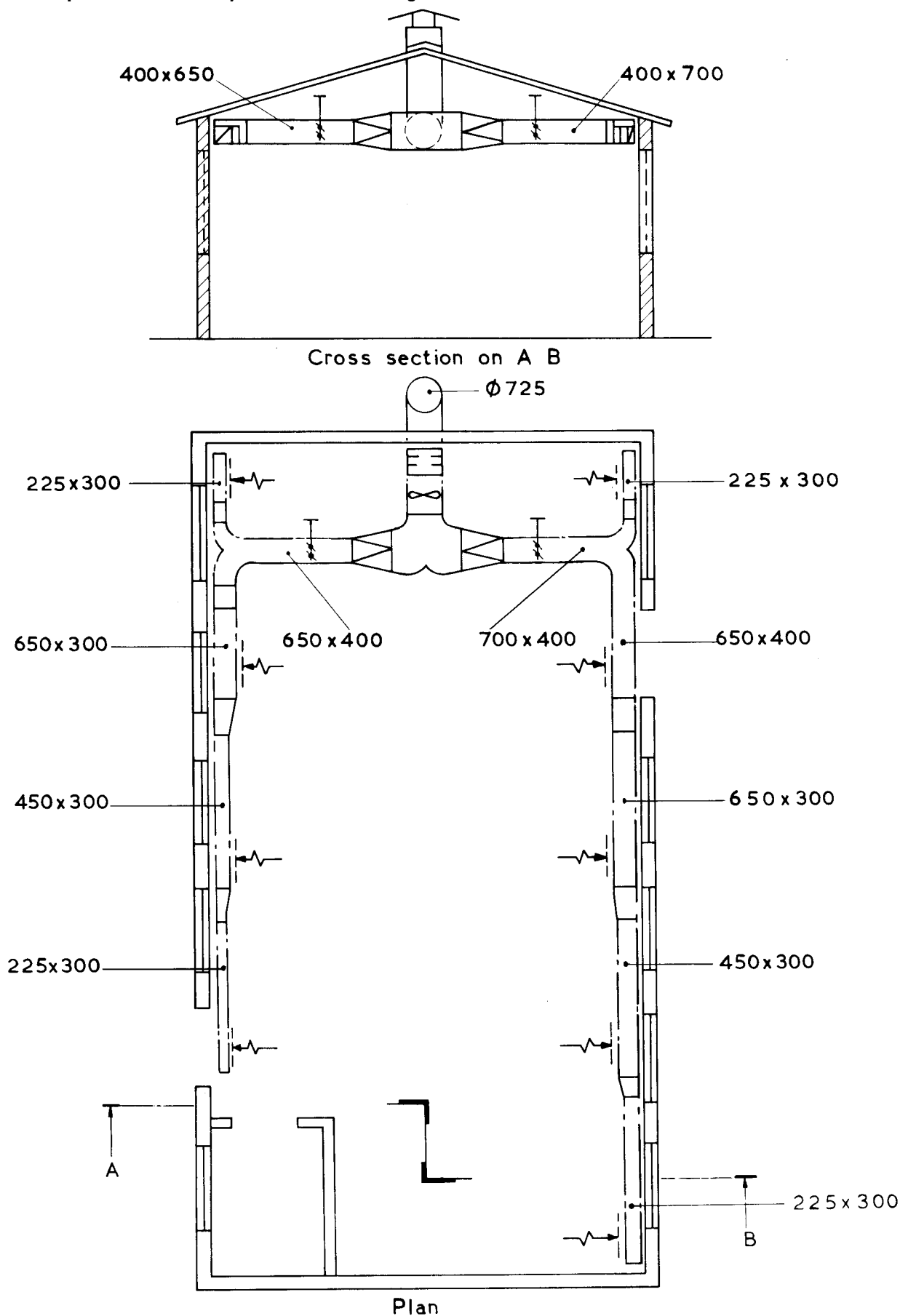


Plan

Symbols are in accordance with BS 1553 : Part 1
All dimensions are in millimetres

Appendix C

Example of the use of symbols on a drawing of an extraction ventilation installation



Symbols are in accordance with BS 1553 : Part 1
All dimensions are in millimetres

Standards publications referred to

- BS 534 Steel pipes, fittings and specials for water, gas and sewage
- BS 1646 Graphical symbols for process measurement and control functions
- BS 3799 Steel pipe fittings, screwed and socket-welding for the petroleum industry
- BS 5070 Drawing practice for engineering diagrams

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The following BSI references relate to the work on this standard:

Committee references MEE/88 and CHE/17

Drafts for comment 71/34910, 71/34911 and 71/35726.

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- British Electrical and Allied Manufacturers' Association
- British Gas Corporation
- British Gear Manufacturers' Association
- British Internal Combustion Engine Manufacturers' Association
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- British Steel Corporation
- British Steel Industry
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- Engineering Equipment Users' Association
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- Health and Safety Executive (HM Factory Inspectorate)
- * Institution of Gas Engineers
- Institution of Heating and Ventilating Engineers
- Institution of Mechanical Engineers
- Institution of Plant Engineers
- Institution of Production Engineers
- London Transport Executive
- Machine Tool Trades' Association

- * Ministry of Defence
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- National Engineering Laboratory – Department of Industry
- Process Plant Association
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- Institution of Civil Engineering
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- Patent Office
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- United Kingdom Atomic Energy Authority

Amendments issued since publication

Amd. No.	Date of issue	Text affected